

Marion Fumani Maphali

Network Engineer · Backend Developer · IT Professional

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PROFESSIONAL SUMMARY

IT Communication Networks graduate from Nelson Mandela University with hands-on experience in enterprise network design, Cisco routing and switching, and Python backend development. Completed a 12-month backend developer internship at DataTank BI Solutions, building data-driven systems, tracking dashboards, and a ticketing web application within a professional BI environment. Delivered iterative development cycles responding to feedback from supervisors, management, and clients to continuously improve system functionality. Holder of multiple Cisco Networking Academy certifications covering networking, cybersecurity, IoT, and enterprise automation. Driven by a passion for continuous learning, technical problem-solving, and building reliable infrastructure.

PROFESSIONAL EXPERIENCE

Python Backend Developer Intern

Jul 2025 – Jul 2026

DataTank BI Solutions · North Riding, Randburg

Hosted by MICTSETA with Lumphindi Business Solutions | 12-Month Internship

- Developed backend software systems using Python for business intelligence applications in a live enterprise environment.
- Built and maintained database interactions and backend business logic for BI data processing pipelines.
- Designed and developed a web application featuring a ticketing system and BI tracking dashboards, iterating functionality based on feedback from supervisors, management, and clients.
- Provided system support and implemented technical solutions across multiple BI projects.
- Collaborated with cross-functional technical teams on software development tasks, debugging, and optimization.
- Applied analytical thinking to design reliable, efficient backend systems aligned with client requirements.

Network Design Engineer (Academic Project)

Jan 2024 – Dec 2024

Nelson Mandela University · Port Elizabeth, South Africa

IT Communication Networks Capstone Project | 12-Month Academic Project

- Designed and implemented a full-scale enterprise airport network infrastructure as part of the final-year capstone project.
- Applied Cisco routing and switching technologies (OSPF, VLANs, BGP, ACLs) across a complex multi-zone topology.
- Implemented VPN, IPSec, NAT/PAT, firewall rules, and network segmentation for security and traffic optimization.
- Produced comprehensive technical documentation, network diagrams, and architecture specifications.
- Simulated and validated the network design using Cisco Packet Tracer and virtual machine environments.

EDUCATION

Diploma in Information Technology: Communication Networks

2021 – 2025

Nelson Mandela University (NMU) · Port Elizabeth

Graduated: April 2025 · Specialization: IT Communication Networks & Enterprise Infrastructure

National Senior Certificate (NSC / Matric)

2020

E.P.P. Mhinga High School · Limpopo

Key subjects: English, Mathematics, Life Sciences, Physical Sciences, Accounting, Xitsonga

CERTIFICATIONS

CCNA: Enterprise Networking, Security & Automation	Cisco NetAcad / NMU	Jan 2025
CCNAv7: Introduction to Networks	Cisco Networking Academy	Dec 2021
Introduction to Cybersecurity	Cisco Networking Academy	Aug 2021
Introduction to IoT	Cisco Networking Academy	Jul 2022
Diploma in IT: Communication Networks	Nelson Mandela University	Apr 2025
National Senior Certificate	Dept. of Basic Education, SA	Nov 2020

TECHNICAL SKILLS

Networking Protocols: TCP/IP, OSI model, IPv4/IPv6, Subnetting, LAN, WAN, VLAN, WLAN

Routing & Switching: Static Routing, OSPF, BGP, EIGRP, VLANs, STP, Cisco IOS

Network Security: Cisco ASA Firewall, VPN (IPSec, Site-to-Site), ACLs, IDS/IPS, NAT/PAT

Tools & Platforms: Cisco Packet Tracer, PuTTY, VirtualBox, Azure (Networking basics)

Programming: Python (backend scripting & automation), C#, SQL

Productivity: Microsoft Office 365 (Word, Excel, PowerPoint, Outlook, Teams)

Troubleshooting: SSH/Telnet, Ping, Traceroute, Wireshark

Operating Systems: Windows (advanced), Linux (CLI basics)

KEY PROJECTS

BI Web Application with Ticketing System & Tracking Dashboards · DataTank BI Solutions

Developed a full-stack web application within a professional BI environment featuring a ticketing system and real-time tracking dashboards. Iterated on features continuously based on feedback from supervisors, management, and end clients, delivering a production-ready solution that supported business intelligence operations.

Airport Network Infrastructure Design & Implementation

Designed a complete enterprise-grade airport network as a capstone project, covering full topology planning, VLAN segmentation, routing protocol configuration (OSPF, BGP), VPN setup, and security hardening across a multi-zone environment using Cisco IOS and Packet Tracer.

Personal Portfolio Website

Designed and built a fully responsive personal portfolio website using HTML5, CSS3, and vanilla JavaScript, featuring animated network node canvas, scroll-triggered transitions, and a contact form. Deployed on GitHub Pages.

ADDITIONAL INFORMATION

Languages: Xitsonga (Native), English (Fluent)

Driver's Licence: Code 10 / Class C1 (Valid until 2030)

Portfolio: <https://marion-mars.github.io/marion-maphali-portfolio/>

Availability: Open to opportunities — immediate start